CMP5367:
Applied Machine Learning

Lecture 1:
Overview of AI and ML

Module Leader: **Dr. Hadeel Saadany**

22nd September 2025

Team

Module Leader : Dr Hadeel Saadany

School of Computing and Digital Technology

Email: Hadeel.saadany@bcu.ac.uk

Linked In Profile:

The Module (House Keeping Rules)- Things to Remember

- If you decide to attend the class, you are expected to be present till the end of the session.
- **Please refrain from leaving room/lab during lecturing (except with permission).**
- Lab sessions will be after the lectures. Labs are the hands-on application of what is learned in the lecture.
- Check your time-table to see which group you are in (please only attend your session).
- Check your Moodle CMP5367 Artificial Intelligence and Machine Learning A S1 2025/6 regularly, for resources and especially for post-class activities.

Support sessions

- You can contact your tutors **Monday to Friday from 9:00 am to 5:00 pm.**
- There will be support sessions announced during the term.
- It is your responsibility to decide if you need to attend the support sessions.
- These are not compulsory but essential for your progress in the module.

Expectations from YOU!

Be proactive in starting your work and seek assistance if needed

Attend all sessions on time and take notes regularly

Behave in a professional manner and avoid disturbing noise in class and lab

Perform all activities to get a full learning experience.

Summative Assessment

➢ 1

Assessment Type **Deliverable1 (D1):** **Coursework: Report**	Level 5	Weighting 80%	Word Count **D1.a** Machine Learning Experiment Portfolio **(60%)** **D1.b** Final Report **(40%)**: 1500-2,000 words (excluding references)
Submission Date D1.a Experiment Portfolio Nov 10th D1.b Report &code Dec 8th	**Submission Time** 3:00 pm	**Module Leader** Hadeel Saadany	

Assessment Information	
Assessment Summary	**D1: Coursework Assessment (80% of total module mark):** There are **two submissions** for this assessment, each contributing to the final report mark: **D1.a**: **Experiment Portfolio** – write-up, insights, and annotated code **(60%)** **D1.b: Final Report & complete codebase (40%)** Includes final report (1500–2000 words) and well-documented, reproducible codebase.

➢ 2

Assessment Type **D2: Presentation (In Person)**	Level 5	Weighting 20%	Word Count 10 Slides
Submission Date Presentation: December 15th	**Submission Time** 03:00 PM	**Module Leader** Hadeel Saadany	**Time Limit (for in person presenting and oral assessments):** 20 minutes

Module Overview

Week number	Lecture	Lab	Submissions
1	AI and ML concepts	Recap and practicals	
2	Data preprocessing	Recap and practicals	**Portfolio Submission 1** **10% of D1**
3	Exploratory Data Analysis (EDA)	Recap and practicals	**Portfolio Submission 2** **20% of D1**
4	ML algorithms: Linear and Logistic Regression	Recap and practicals	
5	ML algorithms: Random Forest & Support Vector Machine	Recap and practicals	
6	Naive Bayes & Model Evaluation	Recap and practicals	
7	Deep Learning	Revision Part I	
8	Feature engineering	Recap and practicals	**Portfolio Submission 3** **30% of D1**
9	Natural Language Processing	Recap and practicals	
10	Hyperparameter Tuning	Recap and practicals	
11	Reinforcement Learning	Revision-Part II	
12	Deliverable Preparations	**Deadline for Assessment 1 Submission**	**Final Report Submission** **40% of D1**

Week 13

Assessment 2 Submission

Deadline December 15th 2025

Group Presentation

Assessment Briefs for Deliverables 1 and 2

- Please read the assessment briefs of the two assessments from the links below:

- Deliverable 1 (D1 Weight 80%)
- Deliverable 2 (D2 Weight 20%)

Learning Outcomes of the Module

Understand machine learning models and train one on real-life data.

Preprocess data for ML training and understand its features.

Identify and evaluate machine learning models in the context of a real-life problem.

Become confident in understanding AI and ML methods, concepts, terminology.

Assessment 1: D1

- **D1.a Machine Learning Experiment Portfolio (60%):**

    1. Dataset Exploration & Research Questions (10% of D1 Mark)
    2. Exploratory Data Analysis (EDA) (20% of D1 Mark)
    3. Predictive Modelling & Evaluation (30% of D1 Mark)

- **D1.b Final Report & Code Submission Group Work (40% of D1 Mark)**

  1500-2,000 words (excluding references)

Objective:

Build and evaluate ML models that address real life problems

- Outline and mark criteria on Moodle under Assessment (Click HERE to access)

!!! READ COURSEWORK BRIEF !!!

Report submission deadline 08th December 2025 @15:00 PM

Good addition to your portfolio to show employers

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Assessment 2: D2

- In-person Presentation (Group Work). 10 Slides. 10 minutes.
- You are expected to answer questions on the experiment you conducted in assessment 1.
- Main objectives:
  1. Technical presentation skill development.
  2. Show that the experiment is your own work and not plagiarised.

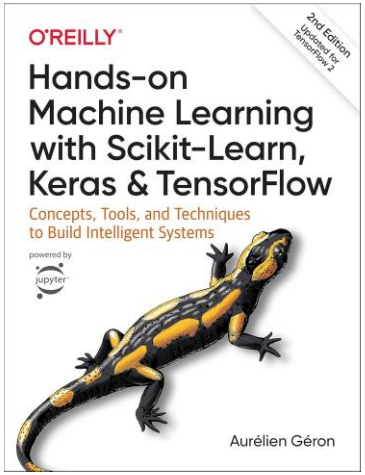
- Outline and mark criteria on Moodle under Assessment (Click HERE to access)

  !!! READ COURSEWORK BRIEF !!!

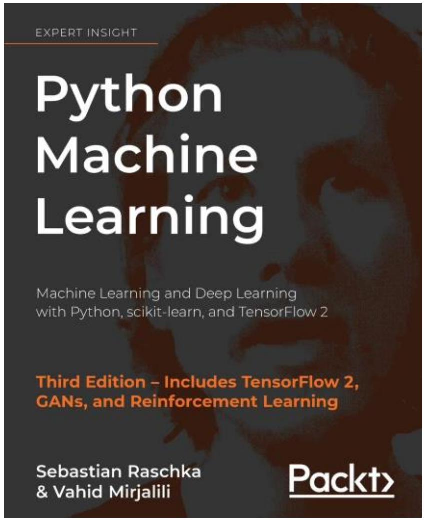
  Deadline:

  15th December 2024

Primary Textbook and Course Resources



Géron, Aurélien, "Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems" (library)

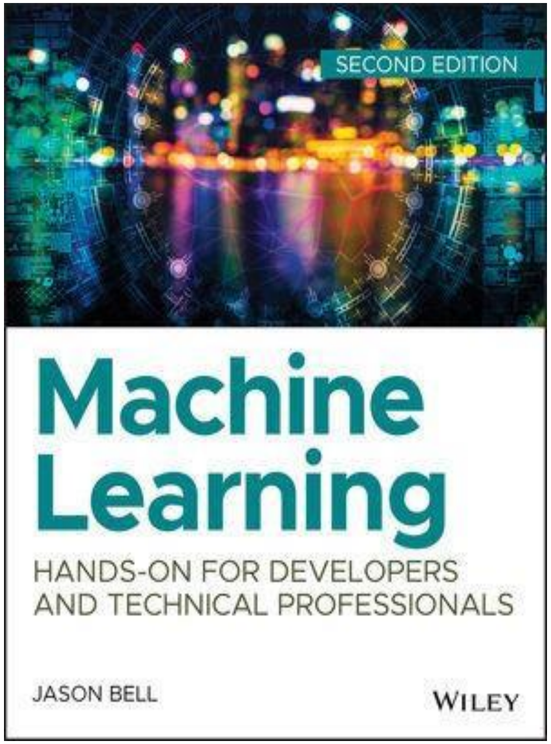


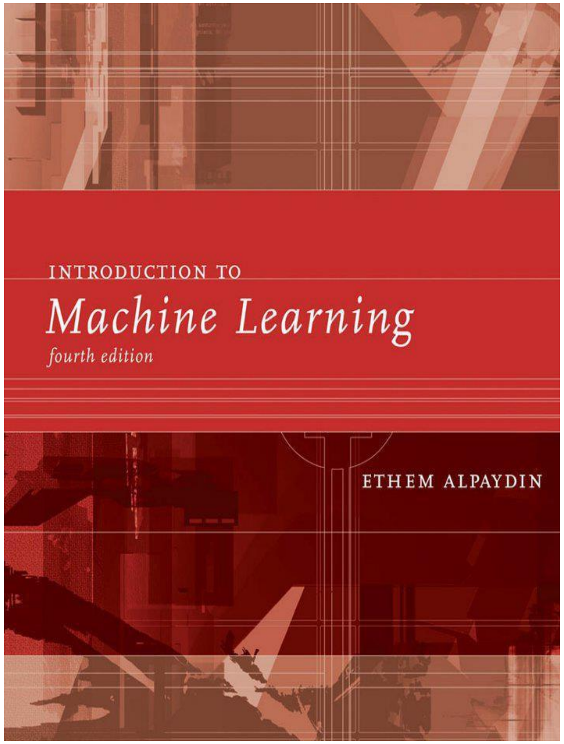
Sebastian Raschka, "Python machine learning : machine learning and deep learning with Python, scikit-learn, and TensorFlow 2. ", (library)

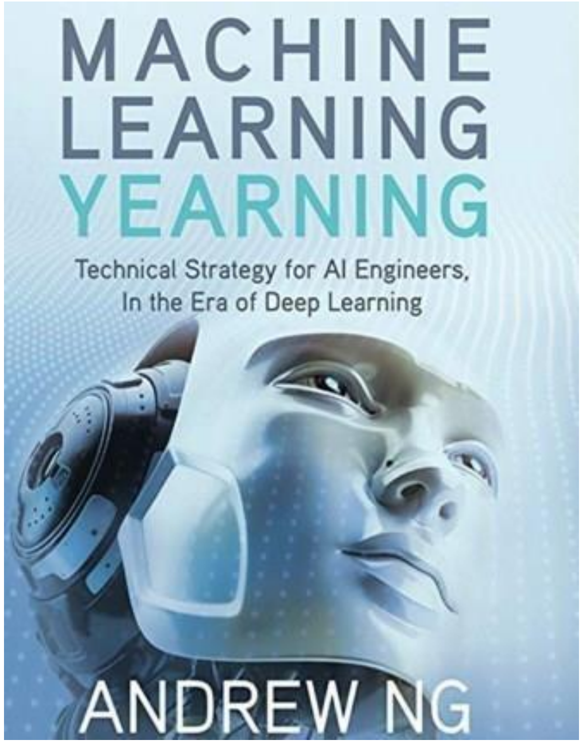
Moodle
Lectures, labs, datasets, cheat sheets, infographics, articles, etc.

- Email and MS Teams Hadeel.saadany@bcu.ac.uk

Additional Book References







Other Resources:

- Ian Goodfellow and Yoshua Bengio and Aaron Courville, "Deep Learning", http://www.deeplearningbook.org/

Any questions so far?

Who are you?
Why are you here?

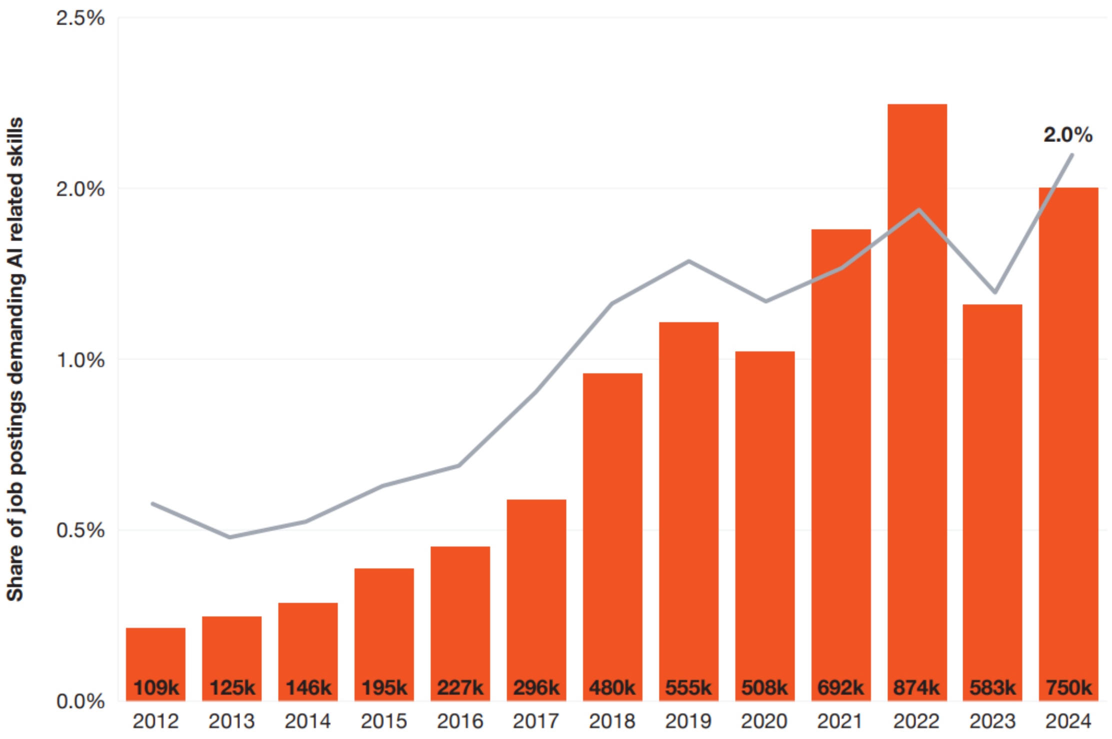
Scan the QR Code

2025 Global AI Jobs Barometer
US Analysis

Core | Aug + Aut | Industry

Despite a weakening labor market in 2024, with fewer job postings overall, demand for roles requiring AI-related skills continues to rise

Total number and share of job postings requiring AI skills, US, 2012-2024



Sources: PwC analysis, Lightcast data

Key findings

- The share of job postings requiring AI-related skills steadily increased year over year from 2012 to 2022.
- This was also the case for the total number of AI jobs, which peaked at 874k in 2022.
- Despite a weaker US job market with fewer roles being posted, AI job postings increased significantly between 2023 and 2024. The share of AI-related jobs increased significantly, this indicates a continued high demand for AI skills.

Notes

- We use Lightcast data for jobs postings, including associated skills.

Source: The Fearless Future: 2025 Global AI Jobs Barometer - US Analysis

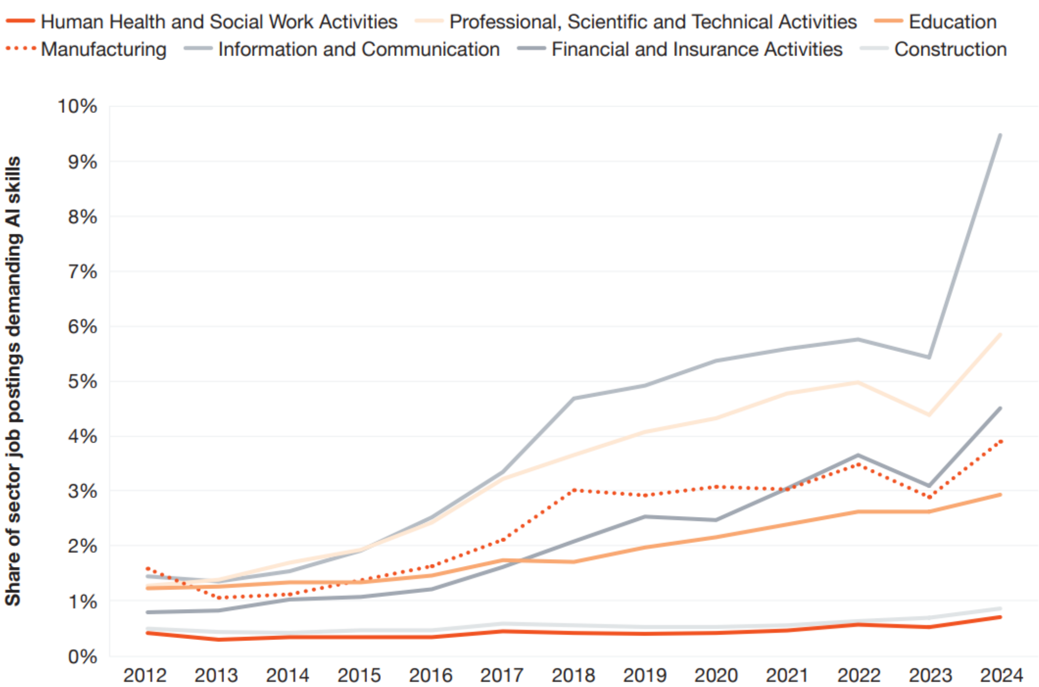
Which job market to target??

2025 Global AI Jobs Barometer
US Analysis

Core | Aug + Aut | Industry

Demand for AI skills has accelerated in all sectors in recent years, with Information and Communications Technology, Professional Services, and Financial Services leading the way

Share of AI job postings by sector, US, 2012-2024



Sources: PwC analysis, Lightcast data

Key findings

- The information and communication sector has experienced the highest growth in demand for AI related skills, rising from 1.4% in 2012 to 9.5% in 2024.
- AI skills requirements are lagging behind in the Construction and Health and Social care sectors, with neither exceeding 1% of job postings with such requirements in 2024.

Notes

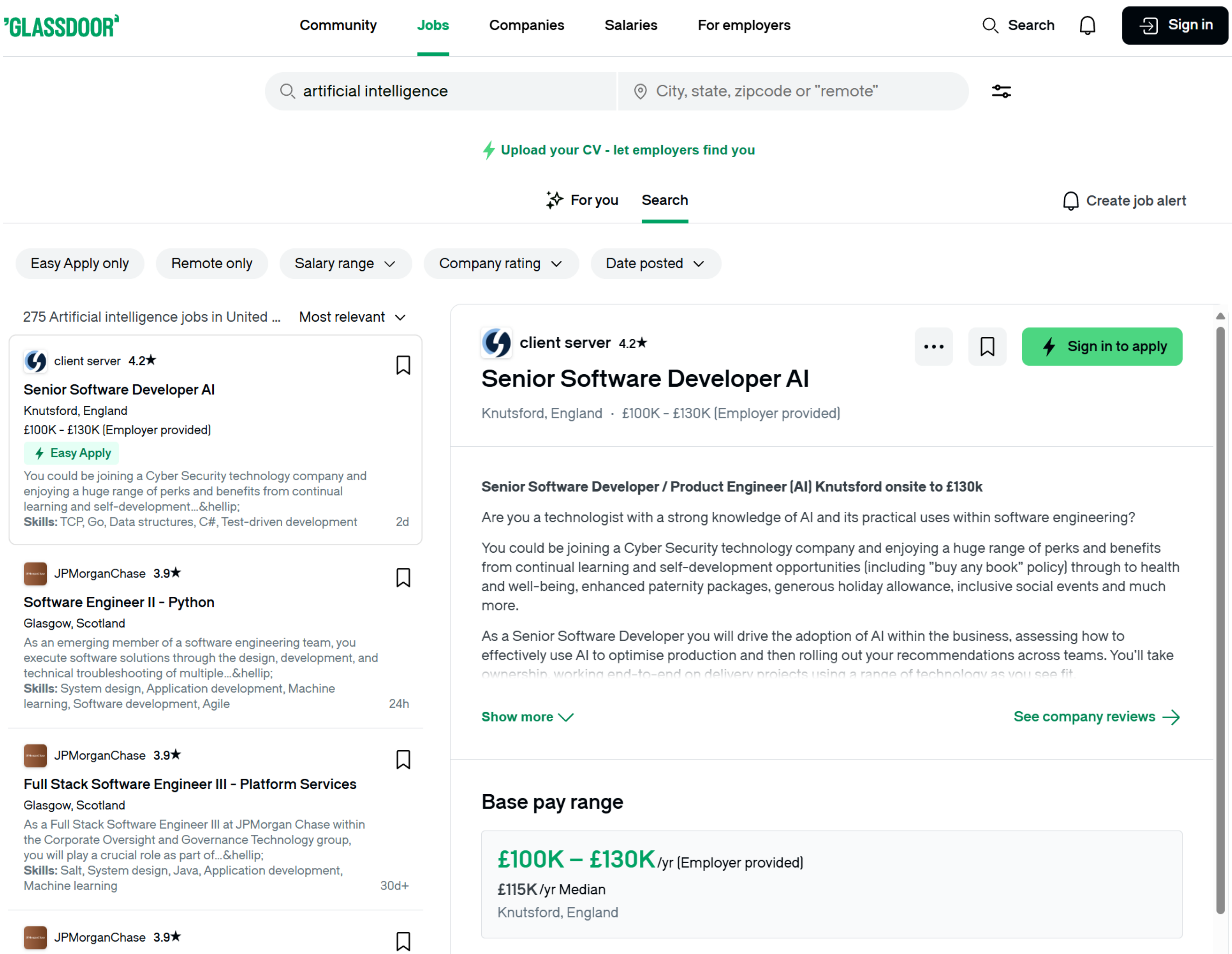
- We use Lightcast data for jobs postings, including associated skills and sectors.

Source: [The Fearless Future: 2025 Global AI Jobs Barometer - US Analysis]

The Most In-Demand AI Jobs of 2025

1. **Machine Learning Engineer**
2. **Data Scientist**
3. **AI Ethicist and Policy Expert**
4. **Robotics Engineer (AI Integration)**
5. **Natural Language Processing (NLP) Specialist**
6. **Computer Vision Engineer**
7. **AI Security Specialist**
8. **AI Research Scientist**

Source: [The Most In-Demand AI Jobs of 2025 | Artificial Intelligence Jobs]



Source: [275 Artificial intelligence jobs in United Kingdom | Glassdoor]

Artificial Intelligence Engineer

Devi Technologies

Birmingham B18 6BA • Hybrid work

£54,000 - £60,000 a year - Permanent, Full-time

Apply now

Job details

Pay

£54,000 - £60,000 a year

Job type

Permanent Full-time

Location

Birmingham B18 6BA • Hybrid work

Full job description

- Understand client requirements and business goals to recommend AI-driven solutions.
- Design, develop, and evaluate AI models (e.g., machine learning, NLP, computer vision).
- Perform data analysis and preprocessing to prepare datasets for AI model training.
- Collaborate with data scientists, engineers, and business stakeholders.
- Stay updated on emerging AI technologies, tools, and best practices.
- Deliver presentations and reports to communicate technical concepts to non-technical audiences.
- Oversee or participate in the deployment and integration of AI solutions.
- Ensure compliance with data privacy, ethical AI use, and regulatory requirements.

Job Types: Full-time, Permanent

Pay: £54,000.00-£60,000.00 per year

Work Location: Hybrid remote in Birmingham B18 6BA

Application deadline: 12/10/2025

Source: https://tinyurl.com/2kkpy4hu

Your assessment 1 = Employability

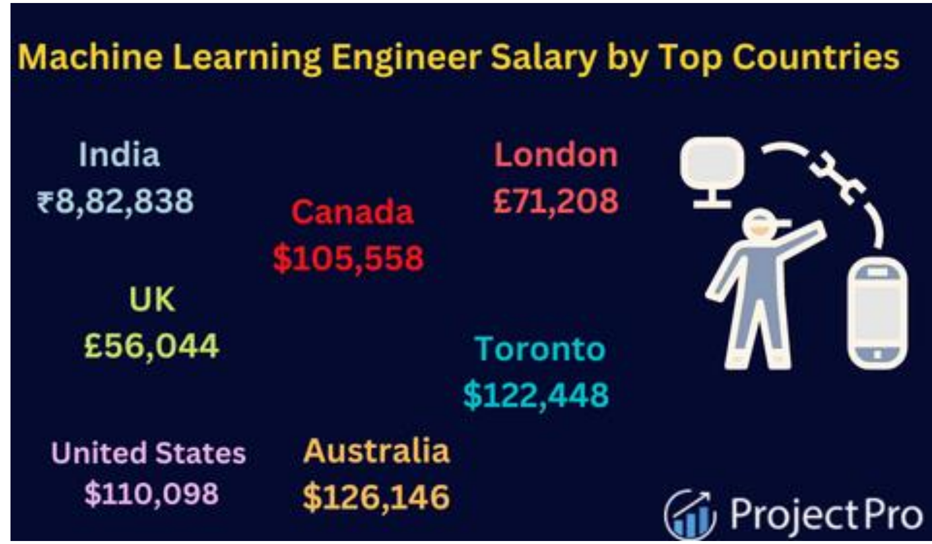
Top 10 jobs involving AI skills

Top jobs seeking machine learning or artificial intelligence skills

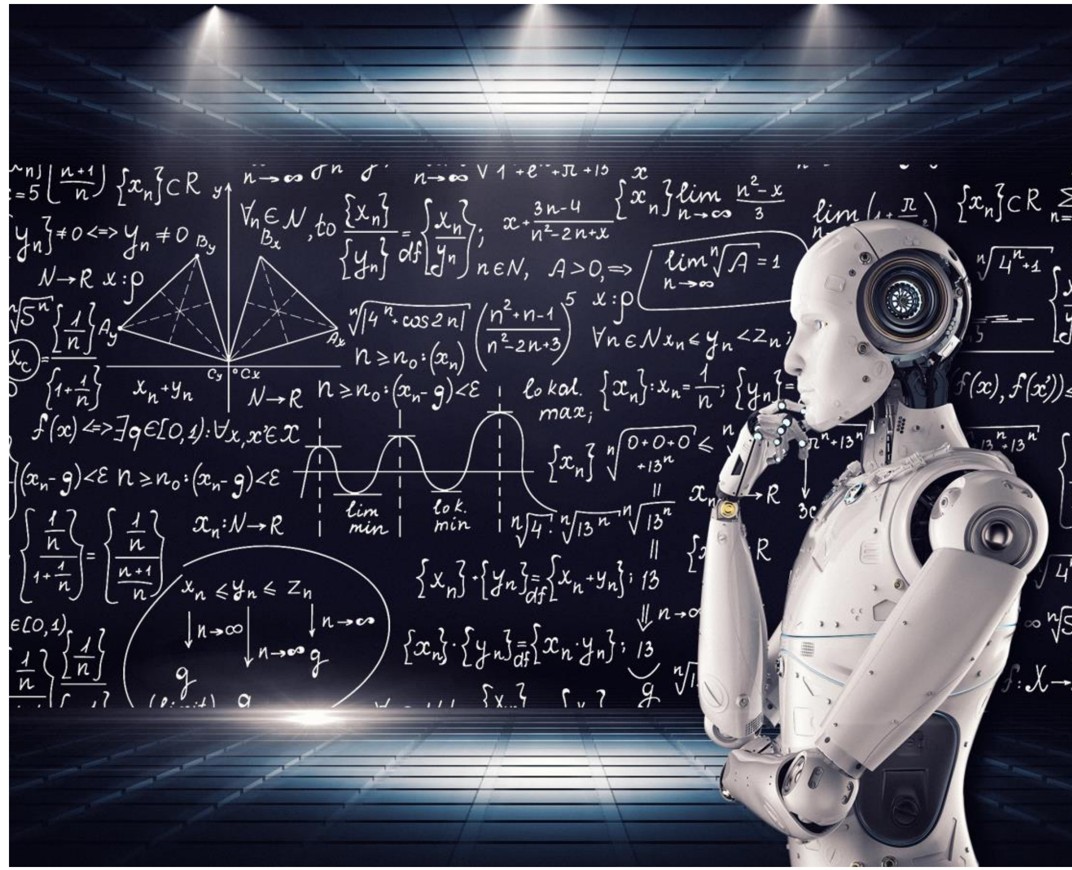
Rank	Job title	% of postings containing AI or machine learning	Rank	Job title	% of postings containing AI or machine learning
1.	Machine learning engineer	75.0%	6.	Algorithm developer	46.9%
2.	Deep learning engineer	60.9%	7.	Junior data scientist	45.7%
3.	Senior data scientist	58.1%	8.	Developer consultant	44.5%
4.	Computer vision engineer	55.2%	9.	Director of data science	41.5%
5.	Data scientist	52.1%	10.	Lead data scientist	32.7%

Source: Indeed

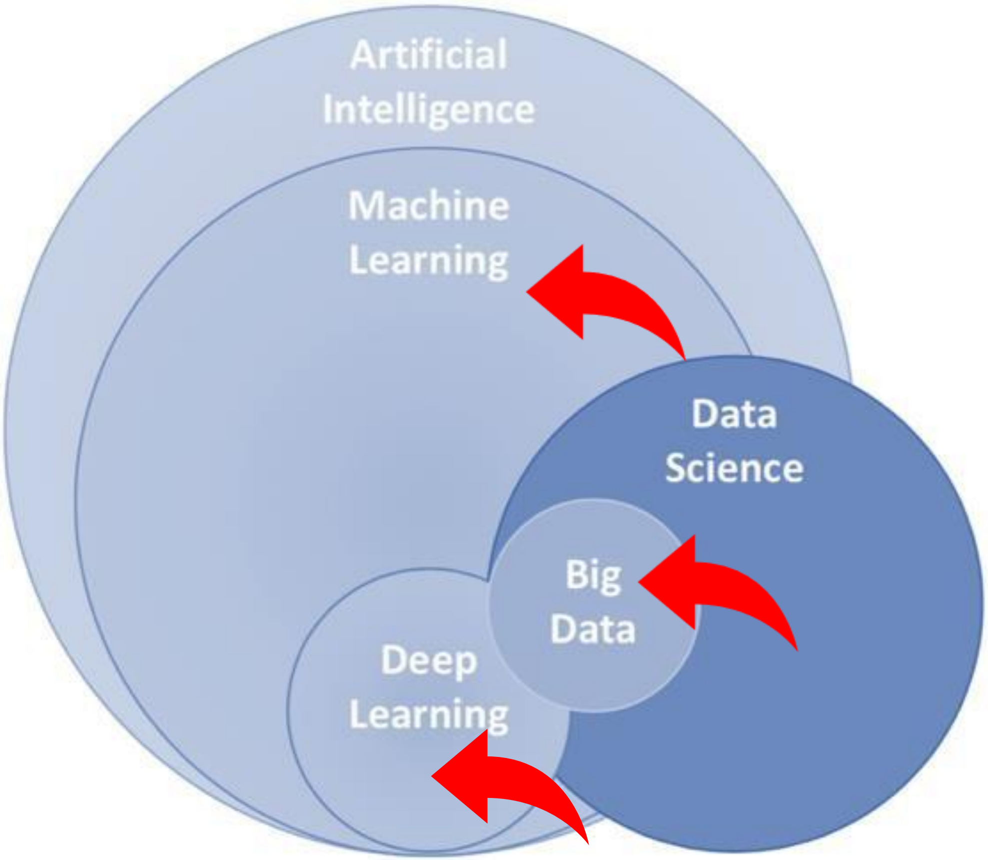
indeed



What is Machine Learning?



AI, ML, Deep Learning, and Big Data



AI, ML, Deep Learning, and Big Data

- **Artificial Intelligence (AI)**

AI refers to technology that trains machines to imitate or simulate human intelligence processes in real-world environments,

- **Machine Learning (ML)**

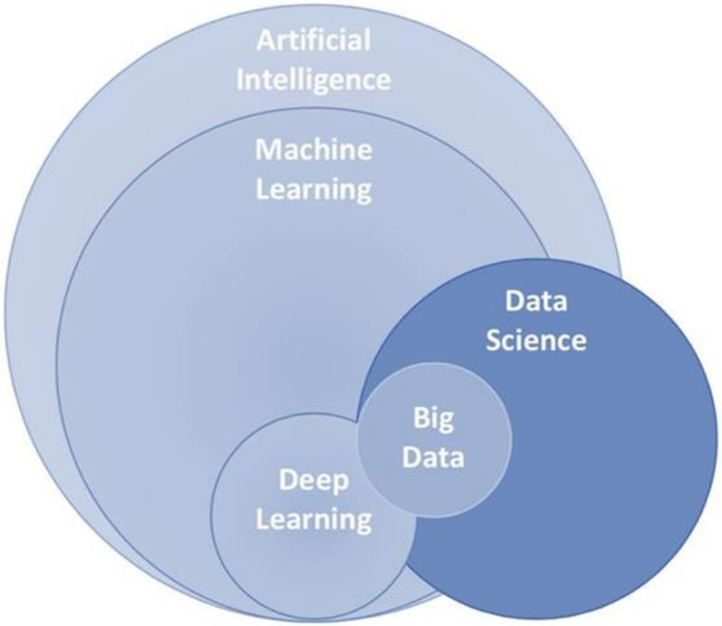
ML is a subset of AI and refers to the process of teaching algorithms to learn patterns from existing data to predict answers on new data.

- **Deep Learning**

Deep Learning is a specialized area of machine learning inspired by the human brain's structure.

- **Big Data Analytics**

Big Data Analytics involves analysing large and diverse datasets to uncover hidden patterns.



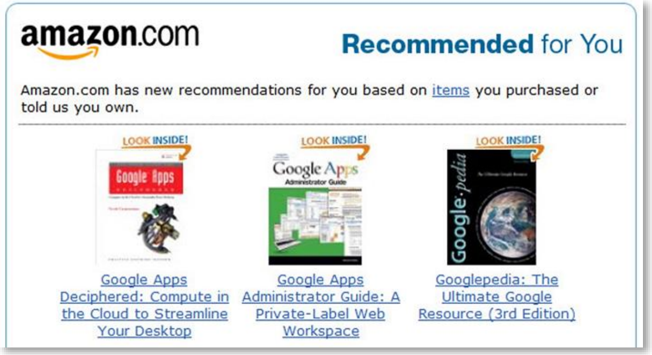
AI and ML use case

Face and voice recognition apps

Filter emails

Keep Internet Safe

Large language models


amazon.com Recommended for You

Amazon.com has new recommendations for you based on items you purchased or told us you own.

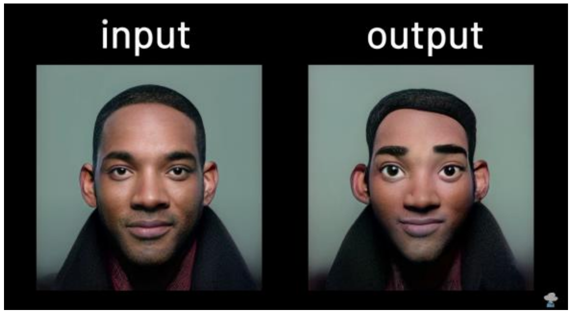
LOOK INSIDE! Google Apps Deciphered: Compute in the Cloud to Streamline Your Desktop

LOOK INSIDE! Google Apps Administrator Guide: A Private-Label Web Workspace

LOOK INSIDE! Googlepedia: The Ultimate Google Resource (3rd Edition)

Personalised customer service

Is AI good or bad??

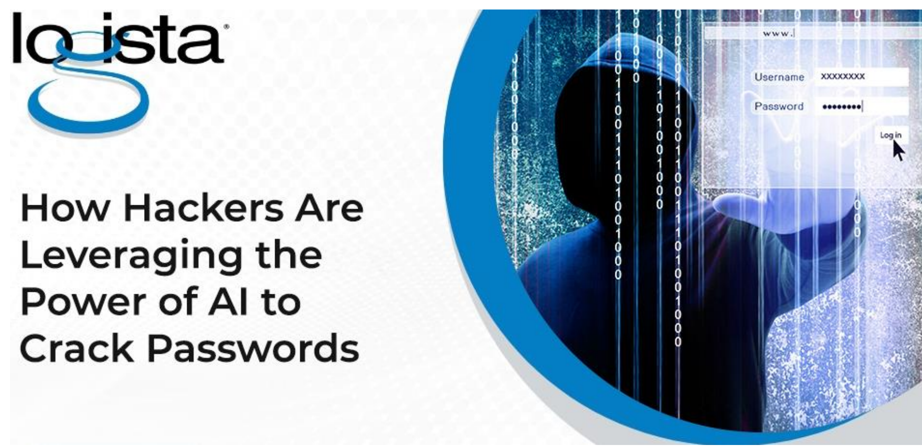


Animations

Medical diagnosis

Self-driving cars

Fake news

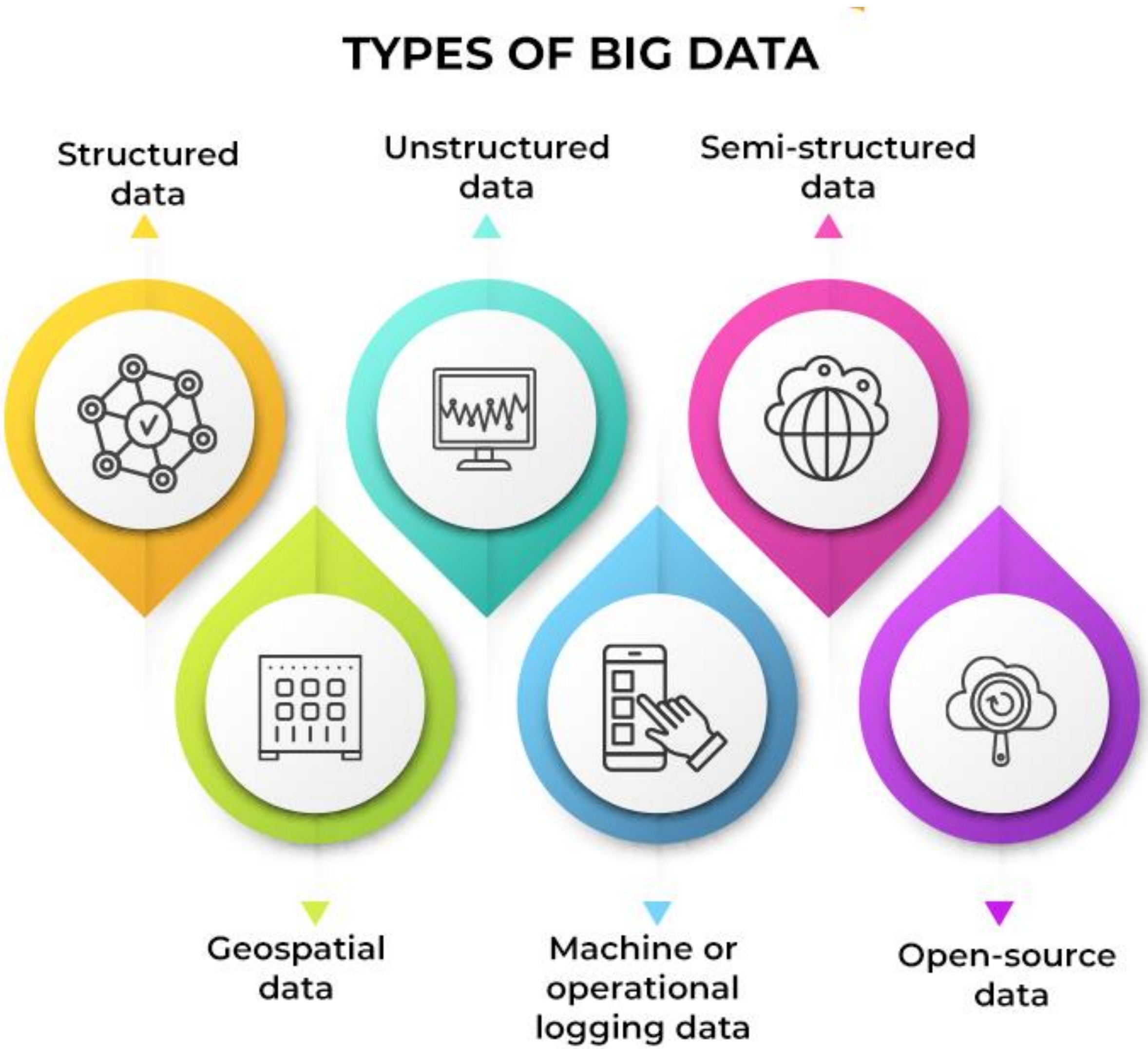


Data and Information

- Society produces huge amounts of data.
- We are concerned with machine learning techniques for automatically discovering patterns in data.
- Can you think of any free online platforms that may be using your data?
- ***There is no such thing as free lunch***

Courtesy: https://datastock.shop/data-is-the-new-oil-of-modern-technology/

What are types of big data?



https://www.spiceworks.com/tech/big-data/articles/what-is-big-data/

	Unstructured Data	Semi-Structured Data	Structured Data
Characteristic	No defined data models; difficult to search	Loosely-coupled data models	Clearly-defined data models; easy to search
Example	Image file	Call center log	Spreadsheet
Storage	Data lake	Organized by metatags	Relational Database

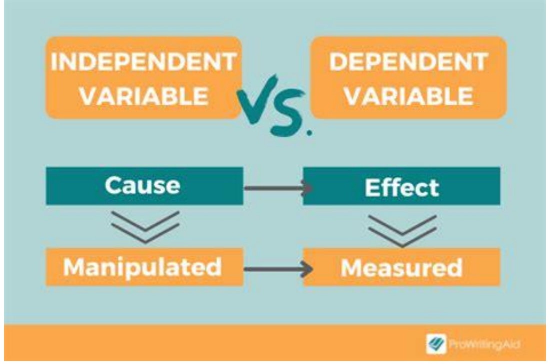
- We are focused on machine learning techniques for automatically uncovering **insights** in data.

Components of ML

- **Each instance** is described by a fixed predefined set of features, its “attributes”.
- Each instance has variables:
- What types of variables:

Dependent and independent

The **independent variable** is the cause. Its value is **independent** of other variables in your study.
The **dependent variable** is the effect. Its value **depends** on changes in the independent variable.



Example: **A scoring system for webpage Phishing Detection**
Your **dependent variable** is the score you are predicting
Your **independent variable** can be the length, the source, etc. If you change it, the score (dependent variable) will also change.

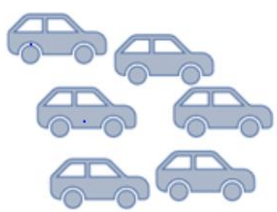
Types of attributes/features/variables

1. Quantitative Data Type:

Anything Which Is Measured By Numbers.

a.) Discrete Data Type:

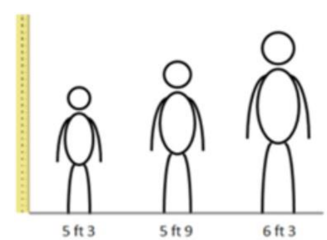
No. of Laptops



No. of Cars

Discrete Values Or Whole Numbers.
e.g.: 20, 13, 1, etc

b.) Continuous Data Type:



Height

Time

Value Within A Certain Range
e.g. 5.77 seconds, 5.772 seconds,

https://pianalytix.com/data-types-in-machine-learning/

Types of attributes/features/variables

2. Qualitative Data Type:

Data Types That Cannot Be Expressed In Numbers.

a. Nominal Data Type

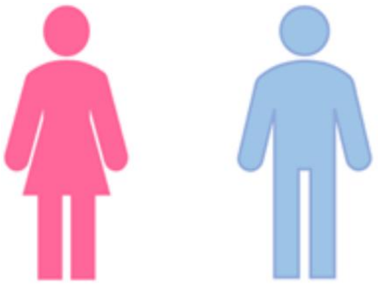
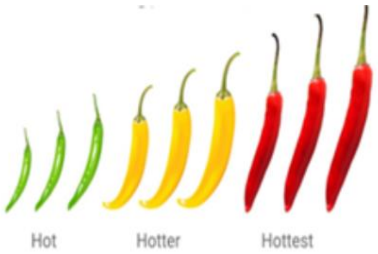


Fig: Gender (Female, Male),

An Example Of Nominal

Names Or Labels: Spam/ham

b. Ordinal Data Type:



Like Nominal Data But Has Some Natural Ordering
e.g.: score findings or risks on the likelihood or impact for cyber security risk.

Types of attributes/features/variables

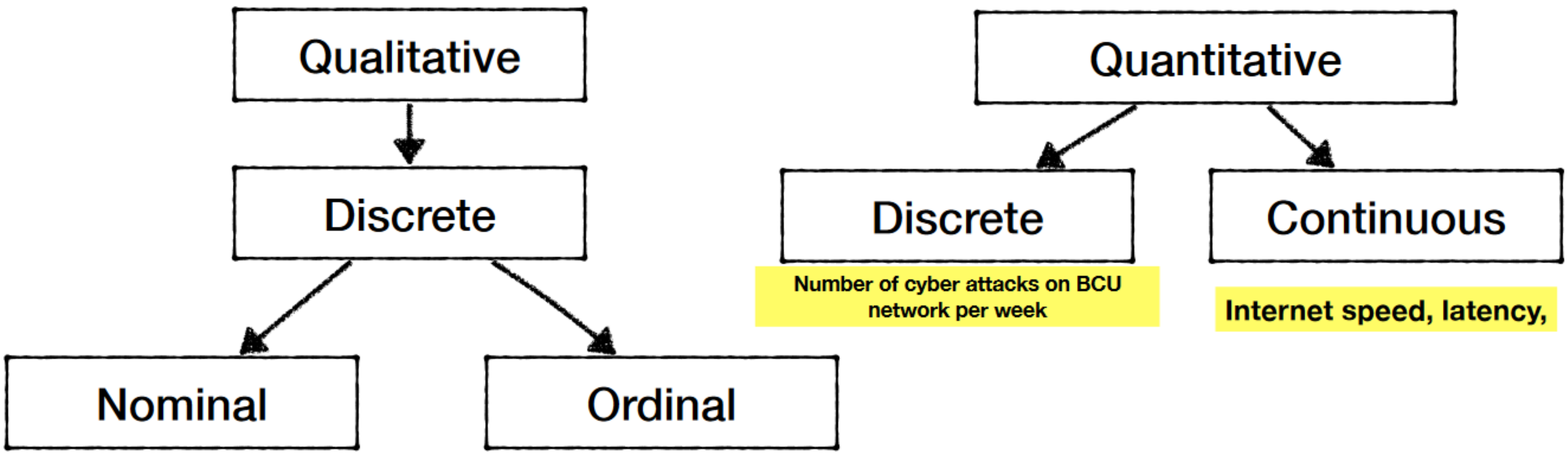
c. Interval Data Type:

Measured along a numerical scale that has equal distances between adjacent values.
e.g. difference between 20 and 21 degrees is identical to the difference between 225 and 226 degrees.

d. Ratio Data Type

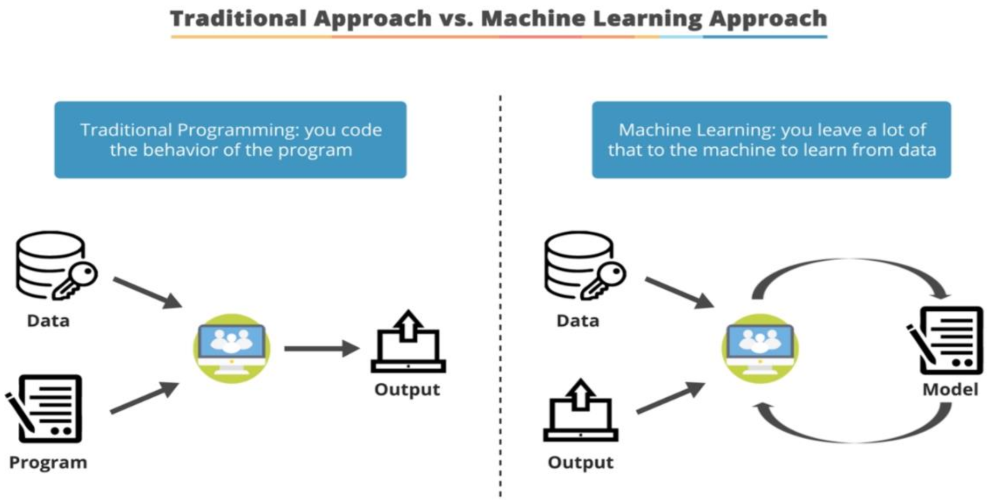
The Same As The Interval Data Type But Has The Absolute Zero. Here Zero Means Complete Absence And The Scale Starts From Zero.

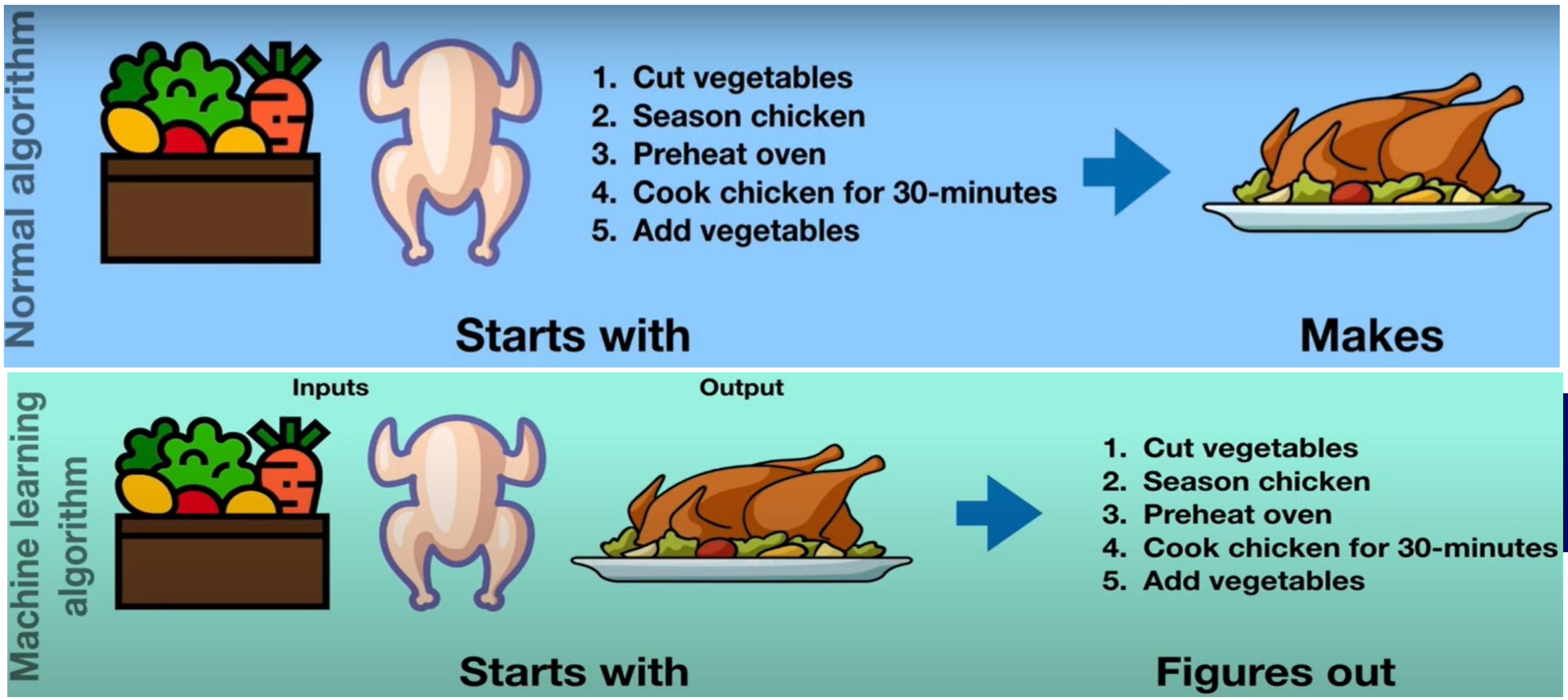
Types of Attributes/ Data



Machine Learning (ML) vs. Traditional Programming

- <u>Traditional Programming:</u> In traditional programming we explicitly specify a set of rules (instructions) for the computer to follow in different scenarios.
- <u>Machine Learning:</u> ML algorithms enable the computers to learn from data, and even improve themselves, without being explicitly programmed.



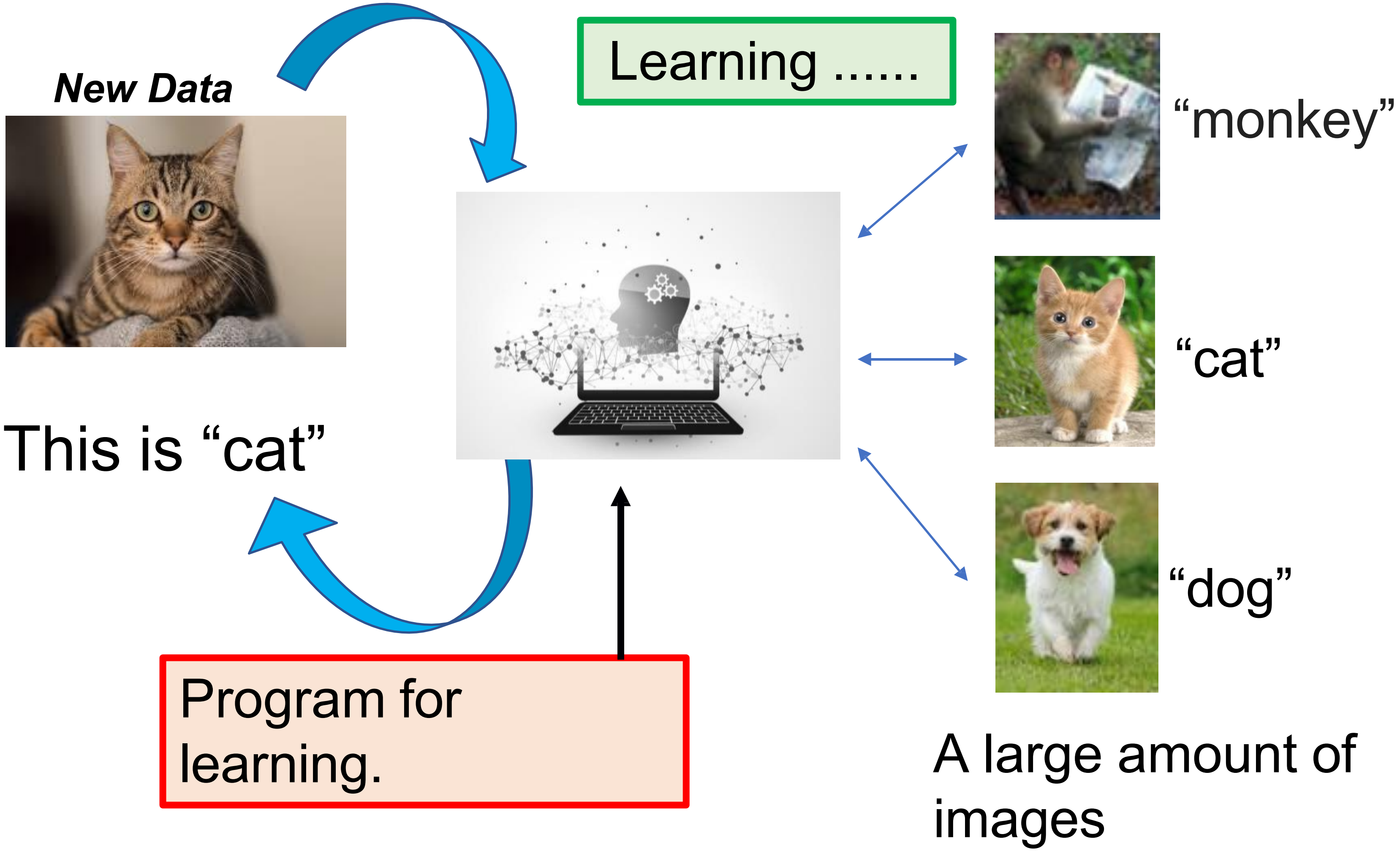


BIRMINGHAM CITY University

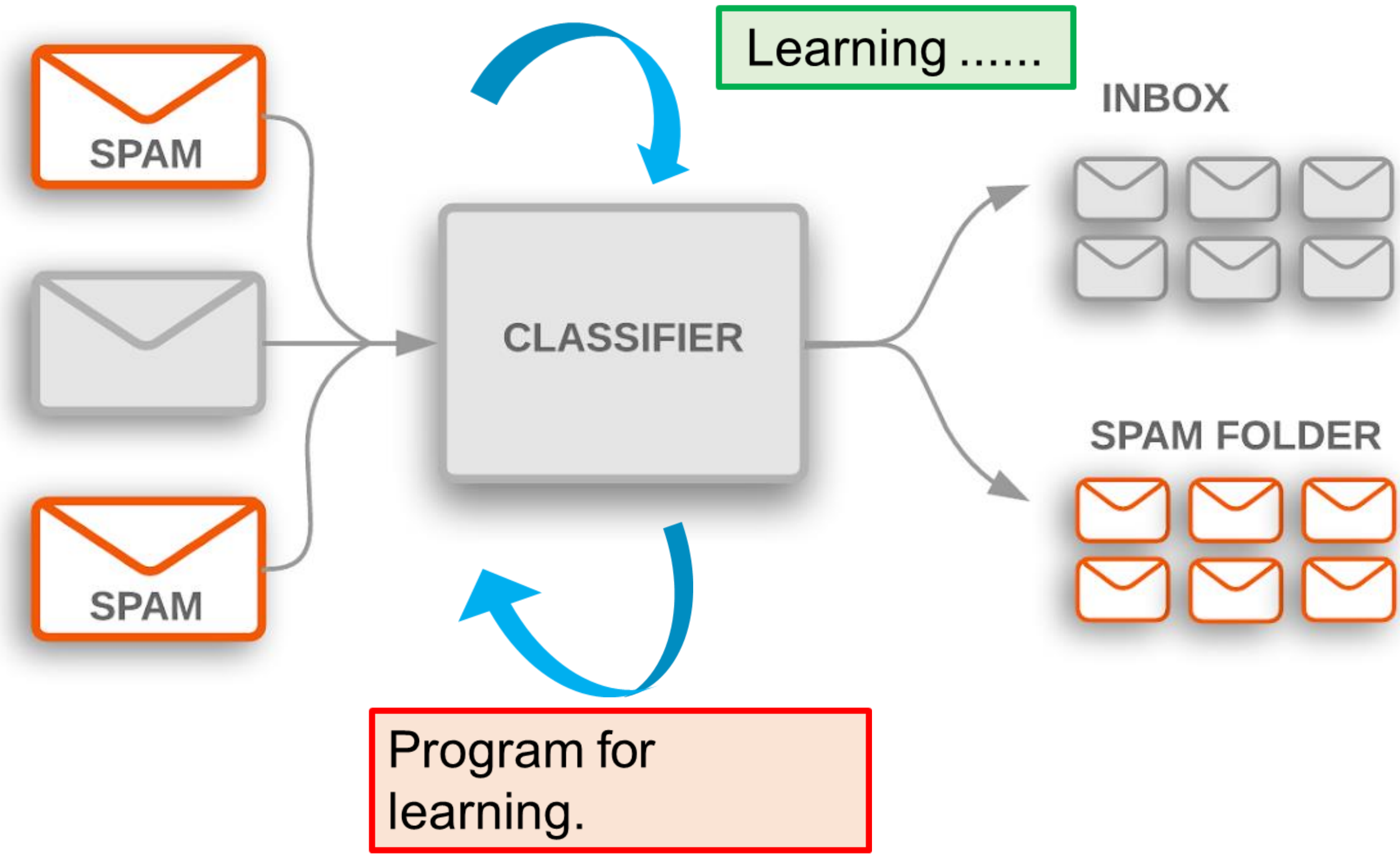
- *The term "Artificial Intelligence" was first coined by **John McCarthy in 1956** during the Dartmouth Conference held at Dartmouth College in New Hampshire.*

- *McCarthy, an American computer scientist, organized the conference with Marvin Minsky, Claude Shannon, and Nathaniel Rochester.*

Image Recognition what can the model learn?

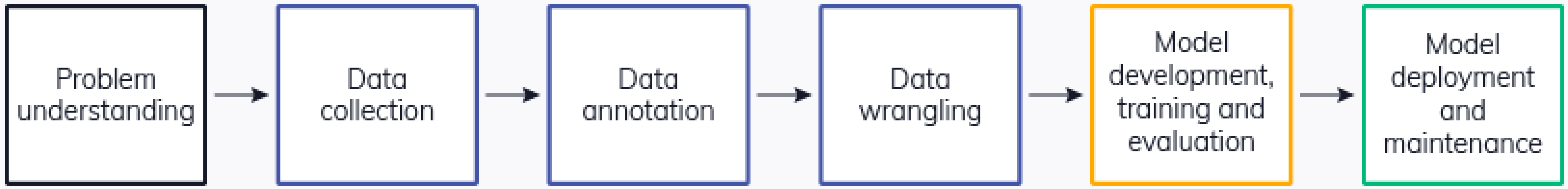


Spam Email detection: What can the model learn?



Machine Learning Lifecycle

- Problem understanding
- Data collection
- Data annotation/Labelling
- Data wrangling/Preprocessing
- Model development, training and evaluation
- Model deployment and maintenance in the production environment



Labelled versus unlabeled data

- Labelled data refers to data that has been manually annotated or tagged with the corresponding correct or desired output.

- Unlabelled data, on the other hand, refers to data that does not have any associated target or class labels. It consists of raw or unlabelled observations without predefined categories or outcomes.

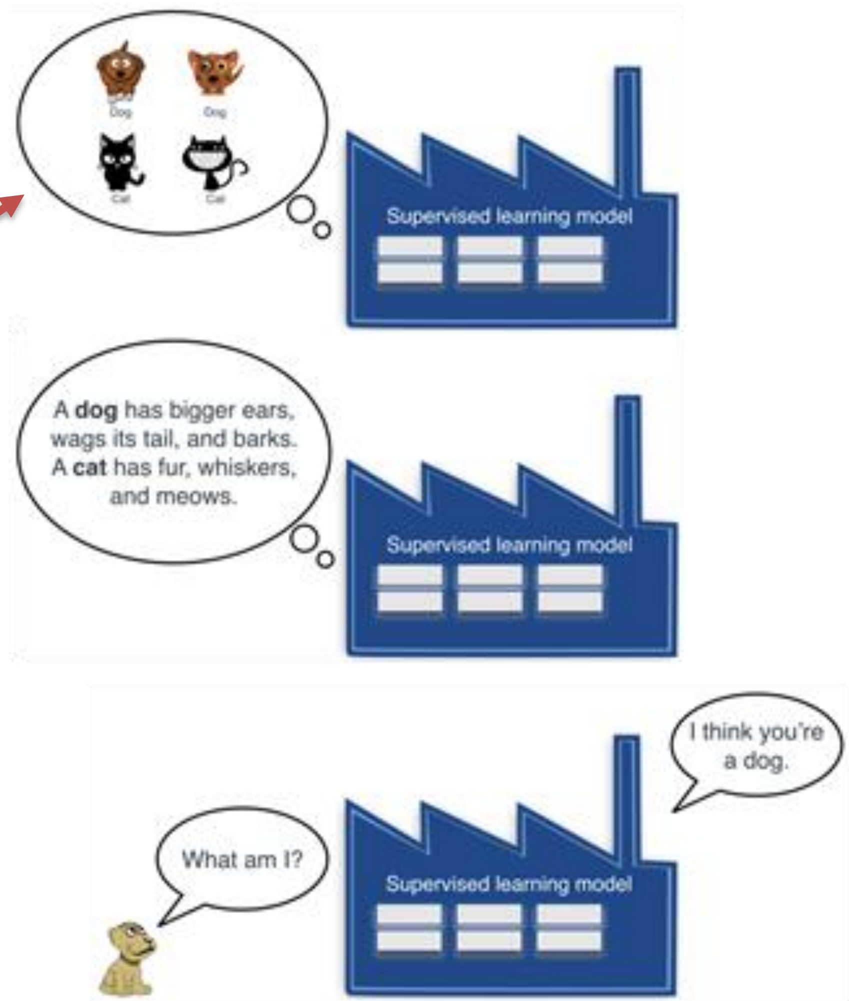
	Label	SMS
0	ham	Later i guess. I needa do mcat study too.
1	ham	But i haf enuff space got like 4 mb...
2	spam	Had your mobile 10 mths? Update to latest Oran...
3	ham	All sounds good. Fingers . Makes it difficult ...
4	ham	All done, all handed in. Don't know if mega sh...

Email	Size	Recipients
1	8	1
2	12	1
3	43	1
4	10	2
5	40	2
6	25	5
7	23	6
8	28	6
9	26	7

Types of Machine Learning

Supervised versus Unsupervised Learning

Supervised



Remember

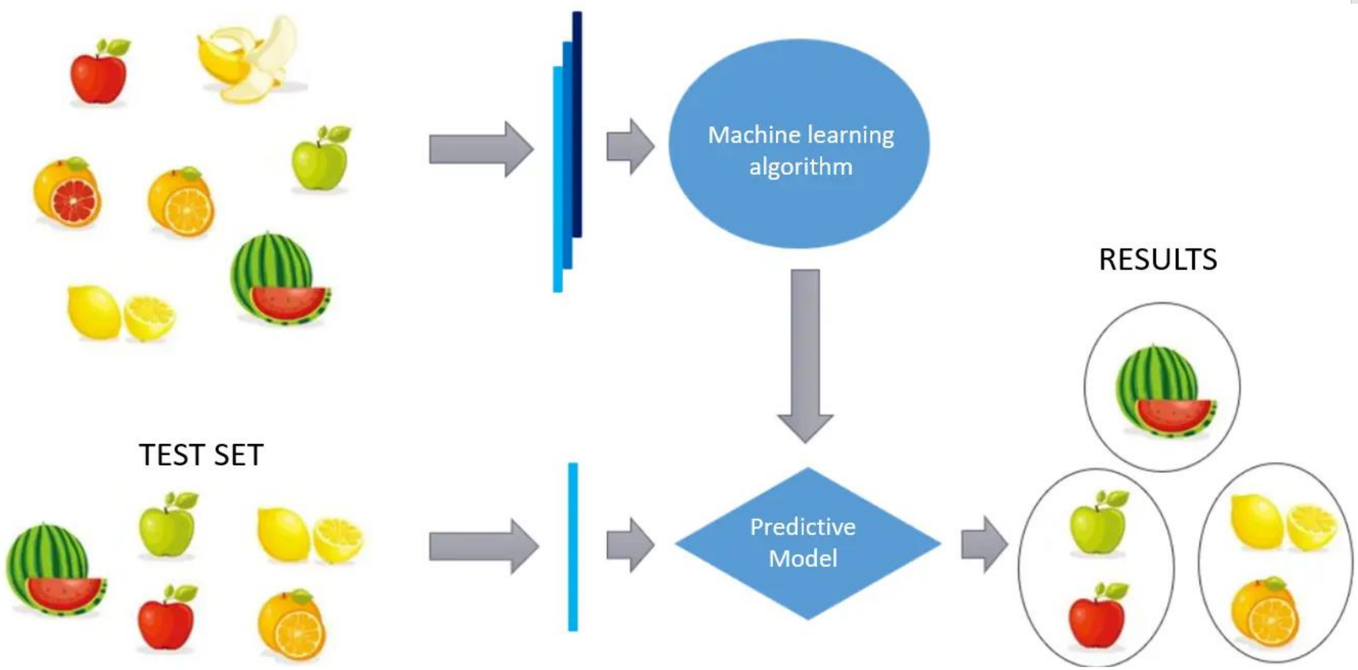
Formulate

Predict

- **Supervised learning** is defined by its use of **labeled** datasets. These datasets are designed to **train** or "**supervise**" algorithms into classifying data or predicting outcomes accurately.

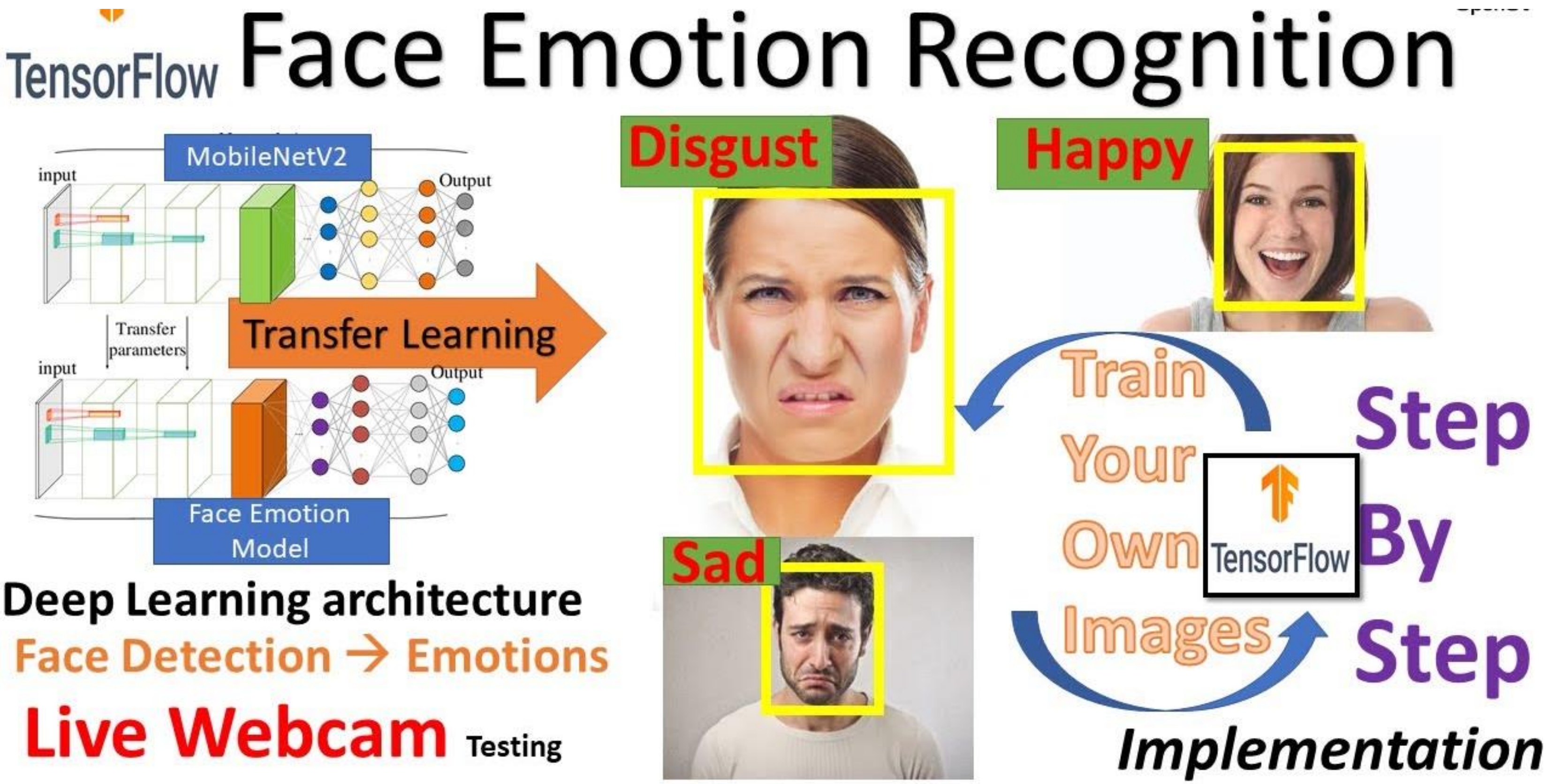
Unsupervised Learning

- Models are given unlabeled data and allowed to discover patterns and insights without any explicit guidance or instruction. **Unsupervised learning** uses algorithms to analyse and cluster **unlabelled** data sets. These algorithms discover **hidden patterns** in data.

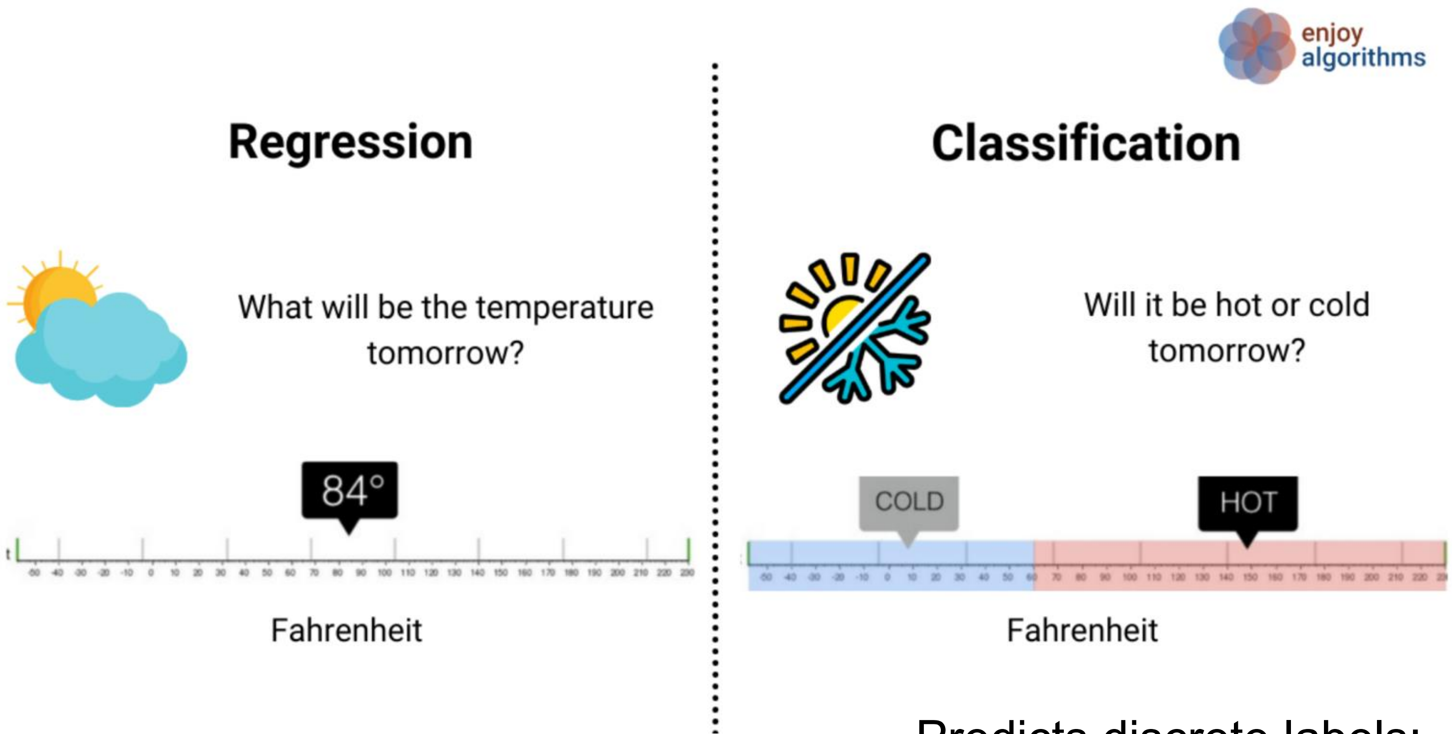


Transferred Learning

Train an image recognition model to recognise face emotions



What are the two types of machine learning prediction?



Prediction of continuous outcomes:
1, 2, 3.5, .56….

Predicts discrete labels:
spam, normal

Illustration of Learning Map – Supervised Learning Temperature

Regression

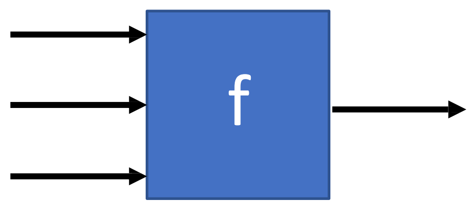
Output of target function f is “scalar”

Predict: Temperature

Today morning 20^0

Yesterday morning 21^0

....



Tomorrow’s morning 22^0

(scalar)

Training Data:

Input:

01/09 morning Temp = 20^0 02/09 morning = 21^0

Output:

03/09 morning Temp = 22^0

Input:

12/09 morning Temp = 30, 13/09 morning Temp = 25

Output:

14/09 morning Temp = 20

Binary Classification

<u>Spam filtering</u>

Function

Yes/No

Yes

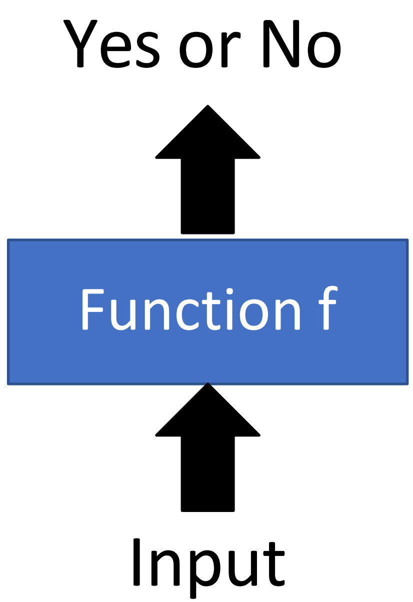
Training Data

No

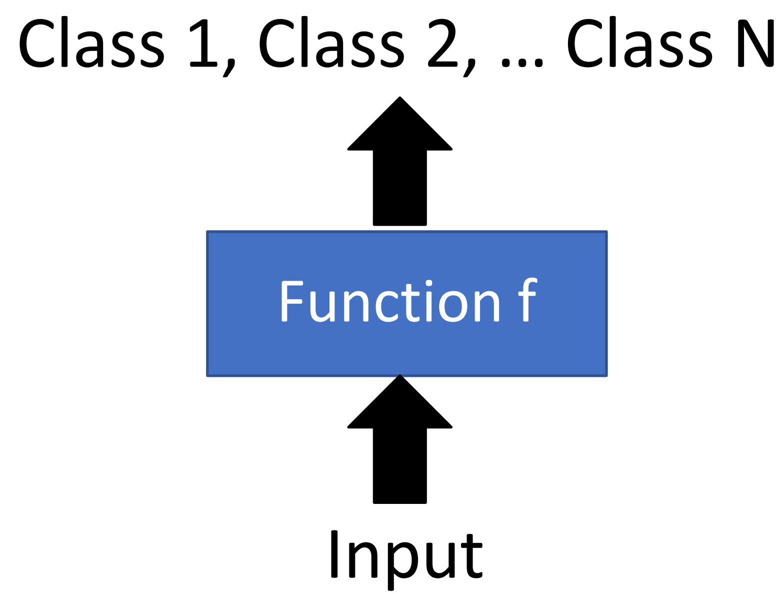
(http://spam-filter-review.toptenreviews.com/)

Illustration of Learning Map – Supervised Learning Classification

- Binary Classification

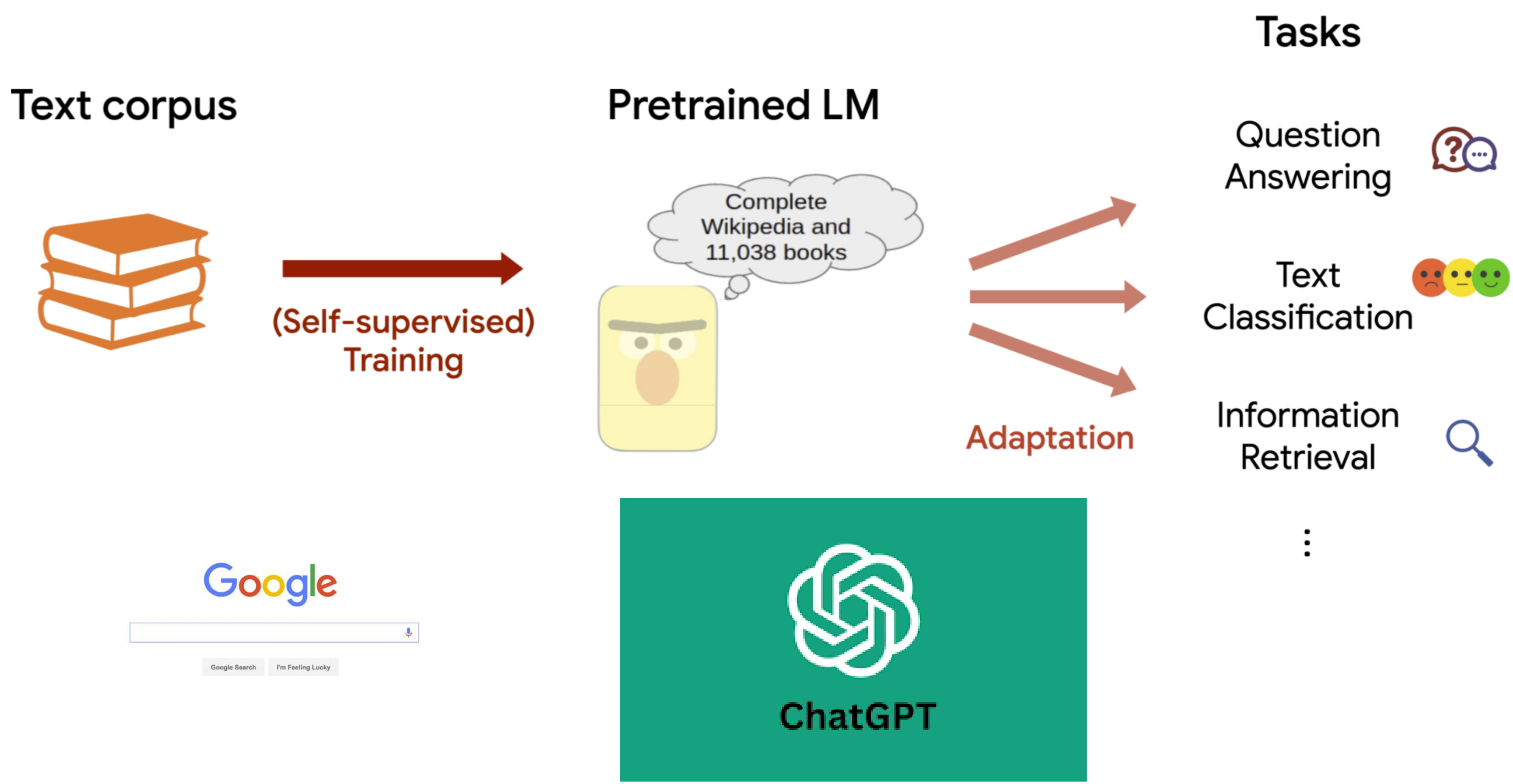


- Multi-class Classification



Self-supervised Learning and Transferred learning

Machine learns the meaning of words from reading a lot of documents



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How do I define the function and how do I tackle the data??

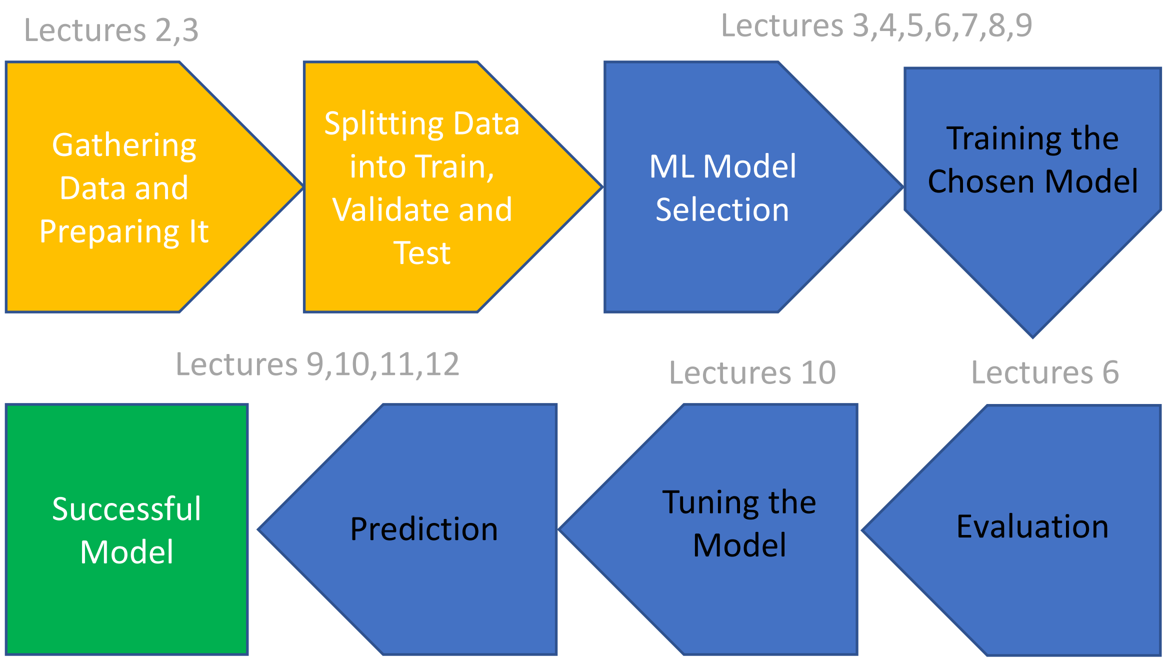
Domain Knowledge

<u>**First step:**</u>

Dig deep into the data and understand it. Apply your domain knowledge or learn a little bit about the domain knowledge of the data to tackle the problem correctly.

My understanding of the data will help me decide which features in the data are important in training a model.

How Machine Learning Works



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Summary of Today's Lecture

What we covered today?

- **Course Introduction**
- **Recap on Machine Learning Concepts**
- **Machine Learning Applications**
- **ML Learning Map**
- **Our course study map.**

Questions?

Thanks for your attention!